Accuracy, Pass@8 and All-8 success by intervention budget (%)

BBH

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Teacher
Mean acc	43.00	42.50	44.75	43.38	47.13	48.88	52.13	53.88	56.00	57.38	62.38	60.63	61.88	63.63	63.88	64.88	65.38	67.88	66.38	69.38	70.13	77.13
Pass@8	79.00	76.00	82.00	78.00	80.00	80.00	84.00	84.00	83.00	85.00	83.00	83.00	84.00	88.00	85.00	83.00	84.00	85.00	86.00	84.00	86.00	89.00
All 8 correct	16.00	6.00	11.00	12.00	13.00	15.00	12.00	18.00	21.00	25.00	30.00	29.00	33.00	37.00	42.00	41.00	43.00	47.00	45.00	47.00	49.00	62.00

CRUXEval-I

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Teacher
Mean acc	21.63	8.50	19.13	33.38	43.25	45.38	47.13	48.38	48.88	49.13	49.25	49.38	49.25	49.25	49.25	49.25	49.25	49.25	49.25	49.38	49.38	50.00
Pass@8	34.00	14.00	32.00	46.00	56.00	55.00	58.00	57.00	57.00	57.00	58.00	58.00	57.00	57.00	57.00	57.00	57.00	57.00	57.00	57.00	57.00	68.00
All 8 correct	7.00	4.00	11.00	24.00	32.00	32.00	35.00	37.00	37.00	38.00	38.00	39.00	39.00	39.00	39.00	39.00	39.00	39.00	39.00	39.00	39.00	34.00

CRUXEval-O

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Teacher
Mean acc	23.25	28.38	38.00	40.75	41.63	42.75	43.63	44.00	43.88	43.75	43.88	43.63	43.63	43.63	43.63	43.63	43.63	43.63	43.63	43.63	43.63	44.88
Pass@8	32.00	40.00	49.00	49.00	49.00	49.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	54.00
All 8 correct	12.00	20.00	27.00	33.00	35.00	37.00	37.00	38.00	38.00	38.00	38.00	38.00	38.00	38.00	38.00	38.00	38.00	38.00	38.00	38.00	38.00	39.00

HumanEval+

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Teacher
Mean acc	64.25	66.00	64.75	68.00	70.25	72.00	73.63	74.75	77.38	78.13	78.50	79.63	79.88	80.00	80.38	80.75	80.75	81.50	81.25	81.25	81.50	81.25
Pass@8	79.00	79.00	78.00	81.00	83.00	85.00	84.00	84.00	84.00	87.00	85.00	85.00	84.00	85.00	86.00	86.00	86.00	86.00	86.00	87.00	87.00	89.00
All 8 correct	45.00	50.00	53.00	56.00	53.00	54.00	58.00	62.00	66.00	67.00	70.00	73.00	73.00	74.00	72.00	74.00	74.00	75.00	75.00	75.00	75.00	69.00

LiveCodeBench

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Teacher
Mean acc	17.25	18.75	20.13	22.13	23.13	23.88	24.25	24.50	24.88	24.63	25.25	25.38	26.25	27.38	27.63	27.63	28.00	28.13	28.50	28.38	28.38	31.00
Pass@8	25.00	24.00	26.00	28.00	30.00	30.00	28.00	30.00	31.00	30.00	30.00	29.00	32.00	32.00	32.00	31.00	34.00	32.00	33.00	34.00	33.00	37.00
All 8 correct	9.00	13.00	13.00	15.00	18.00	18.00	19.00	21.00	19.00	19.00	20.00	20.00	22.00	24.00	24.00	24.00	25.00	26.00	25.00	25.00	26.00	26.00

MATH500

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Teacher
Mean acc	76.00	76.38	75.13	76.13	76.00	77.00	78.00	79.75	81.25	82.50	82.13	81.88	82.63	83.00	84.00	83.88	85.13	85.00	84.75	85.38	85.75	87.75
Pass@8	92.00	91.00	92.00	93.00	92.00	88.00	92.00	93.00	93.00	92.00	94.00	93.00	93.00	93.00	94.00	92.00	94.00	95.00	95.00	95.00	95.00	94.00
All 8 correct	53.00	51.00	50.00	56.00	55.00	58.00	58.00	63.00	64.00	66.00	64.00	66.00	66.00	64.00	70.00	67.00	69.00	66.00	65.00	69.00	68.00	72.00

MBPP+

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Teacher
Mean acc	50.00	55.25	58.75	60.50	63.00	65.13	67.00	69.25	69.25	70.50	70.38	70.38	71.13	71.00	71.25	72.00	71.75	72.13	72.13	72.13	72.38	72.50
Pass@8	61.00	67.00	69.00	67.00	71.00	71.00	75.00	76.00	76.00	76.00	76.00	76.00	76.00	76.00	76.00	77.00	76.00	77.00	77.00	77.00	77.00	79.00
All 8 correct	39.00	42.00	48.00	52.00	55.00	56.00	56.00	62.00	61.00	63.00	65.00	64.00	67.00	65.00	65.00	66.00	67.00	68.00	68.00	68.00	68.00	65.00

OlympiadBench

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Teacher
Mean acc	45.25	48.75	47.13	47.88	48.88	51.00	51.63	51.63	53.38	53.88	53.88	55.13	55.63	55.25	57.63	56.63	56.88	58.38	58.38	58.75	58.63	63.50
Pass@8	70.00	70.00	71.00	70.00	70.00	69.00	75.00	69.00	74.00	71.00	75.00	76.00	74.00	73.00	78.00	76.00	75.00	75.00	76.00	76.00	77.00	82.00
All 8 correct	23.00	23.00	23.00	24.00	26.00	31.00	29.00	34.00	34.00	35.00	34.00	38.00	39.00	39.00	39.00	38.00	37.00	42.00	42.00	43.00	43.00	40.00

Overall

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Teacher
Mean acc	42.58	43.06	45.97	49.02	51.66	53.25	54.67	55.77	56.86	57.48	58.20	58.25	58.78	59.14	59.70	59.83	60.09	60.73	60.53	61.03	61.22	63.50
Pass@8	59.00	57.63	62.38	64.00	66.38	65.88	68.25	67.88	68.50	68.50	68.88	68.75	68.75	69.25	69.75	69.00	69.50	69.63	70.00	70.00	70.25	74.00
All 8 correct	25.50	26.13	29.50	34.00	35.88	37.63	38.00	41.88	42.50	43.88	44.88	45.88	47.13	47.50	48.63	48.38	49.00	50.13	49.63	50.50	50.75	50.88

Shared color scale: 0% (light) to 100% (dark). 100 problems/dataset × 8 rollouts. Teacher starts from the original prompt; intervention runs share a frozen student prefix.